

Syllabus Spring 2025

EMCH 770

Sensitivity Analysis, Uncertainty Quantification, and Predictive Modeling: Combining Experiments with Computations

Instructor: Ruixian Fang; FANGR@cec.sc.edu

Meeting Time/Place 2025:

Office Hours: T/Th, 10AM-11:45AM; Rm A114, 300 Main St

Recommended Texts:

Ref. 1. D.G. Cacuci, *Sensitivity and Uncertainty Analysis: Theory*, Volume 1, Chapman & Hall/CRC, Boca Raton, 2003.

Ref. 2. D.G. Cacuci, *BERRU Predictive Modelling: Best Estimate Results with Reduced Uncertainties*, Springer, 2019; ISBN 978-3-662-58395-1;
<https://doi.org/10.1007/978-3-662-58395-1>.

Catalog Course Description: EMCH 770: Definition and sources of experimental and computational uncertainties; combining imprecisely known experiments with computations and their uncertainties to obtain “best-estimate” results with reduced uncertainties; predictive modeling.

Course Objectives:

1. Comprehend and compute sensitivities of model results to model parameters, initial and boundary conditions.
2. Comprehend and compute uncertainties (variances and covariances) of model parameters.
3. Comprehend and compute uncertainties (variances and covariances) of model responses (results).
4. Combine experiments and computations, including associated uncertainties to predict best-estimate results (model parameters and responses) with reduced uncertainties.

Prerequisites: Matrix algebra; Ordinary linear and nonlinear differential and integral equations; Linear partial differential equations; Green’s functions; Adjoint operators; Probability theory; Statistics.

<u>References</u>	<u>Subject</u>	<u>Weeks</u>
Ref. 2	Elements of predictive modeling	1
Ref. 1	Concepts of Matrix and Operator Theory	1
Ref. 1	Concepts of Probability Theory	1
Ref. 1	Measurement Errors and Uncertainties	1
Ref. 1	Sens. & Uncertainty Analysis of Linear Systems	2
Ref. 1	Sens. & Uncertainty of Nonlinear Systems	2
Ref. 2	Model Validation	1
Ref. 2	Experimental Data Integration	2
Ref. 2	Model Calibration	1
Ref. 2	Best-Estimate Predictions	2

*Individual Projects Assigned: 1/XX/2025; Part 1 due: 2/XX/2025; Part 2 due: 3/XX/2025; Part 3 due: 3/XX/2025; Part 4 due: 4/XX/2025. Progress towards completion of the individual projects will be monitored and discussed individually, as needed. The course grade will be based on the cumulative successful %-completion of individual projects. Grading scale: A (above 90%); B (80% – 89%); C (70% – 79%); D (60% – 69%); F (below 60%).